

DTQEM: A Phenomenological Model for Correlated Path-Information and Thermal Decoherence (v18.0-C)

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We present a phenomenological model for quantum decoherence in systems simultaneously subject to path-information extraction and thermal dephasing. The coherence function $C = C_0 \exp(-aI_{\text{path}} - b\Delta T/T_{\text{ref}} - cI_{\text{path}}\Delta T/T_{\text{ref}})$ extends the independent-bath baseline (v17.0-C, $c = 0$) by a bilinear cross-coupling term c , capturing the synergistic action of correlated environmental modes. Calibrated at $T_{\text{ref}} = 300$ K, the conservative ($c = 0$) limit achieves $R^2 = 0.9679$ and LOOCV $R^2 = 0.9356$; the full model reaches $R^2 = 0.9982$ (LOOCV = 0.9814) on the same synthetic dataset. Re-fitted to the eight-point dataset of Hornberger et al. (2003), the functional form yields $R^2 = 0.936$, demonstrating compatibility with molecule-interferometry data. A second-order cumulant analysis suggests strong inter-bath correlations, and an exploratory numerical scan reveals a thermal crossover at $T^* \approx 3.82$ K, flagged as a preliminary observation requiring low-temperature experimental confirmation. All code is openly available. A companion paper (Paper II) will provide the full microscopic derivation.

I. INTRODUCTION

Quantum decoherence—the irreversible loss of phase coherence in quantum superpositions—limits every practical quantum technology [1, 2]. Two physically distinct channels often coexist: (i) which-path information leakage, reducing fringe visibility via the Englert–Greenberger–Yasin (EGY) relation [3]; and (ii) thermal dephasing from stochastic phase kicks.

In the simplest treatment, these channels add independently:

$$C_{\text{v17}} = C_0 \exp\left(-aI_{\text{path}} - b\frac{\Delta T}{T_{\text{ref}}}\right), \quad (1)$$

where $I_{\text{path}} \in [0, 1]$ is the EGY distinguishability. However, in molecular and solid-state platforms, the same phonon bath often mediates both channels, producing a cross term that Eq. (1) cannot capture. We formulate a minimal extension—DTQEM v18.0-C—adding a bilinear coupling $cI_{\text{path}}\Delta T/T_{\text{ref}}$, calibrate it, and test it against independent data. This paper focuses on the phenomenological model; a companion paper (Paper II) will provide the full microscopic derivation from the Feynman–Vernon influence functional.

II. MODEL DEFINITION

A. Core equation

$$C(I_{\text{path}}, T) = C_0 \exp\left(-aI_{\text{path}} - b\frac{\Delta T}{T_{\text{ref}}} - cI_{\text{path}}\frac{\Delta T}{T_{\text{ref}}}\right), \quad (2)$$

with $\Delta T = T - T_{\text{ref}}$, $T_{\text{ref}} = 300$ K fixed, and four free parameters $\{C_0, a, b, c\}$. Setting $c = 0$ recovers Eq. (1).

B. Physical interpretation

- aI_{path} : EGY path-dephasing, consistent with Caldeira–Leggett weak coupling [2].
- $b\Delta T/T_{\text{ref}}$: linear thermal dephasing from the high- T Ohmic expansion $\coth(\hbar\omega/2k_B T) \approx 2k_B T/\hbar\omega$.
- $cI_{\text{path}}\Delta T/T_{\text{ref}}$: phenomenological cross-coupling for shared bath modes. Motivated by physical boundary conditions and a second-order cumulant argument (Appendix A); treated as a fit parameter.

C. Operational mapping of I_{path}

In circuit-QED, the mean intracavity photon number $\bar{n}$ gives [5]

$$I_{\text{path}}(\bar{n}) = \sqrt{1 - e^{-\kappa\bar{n}}}. \quad (3)$$

This is a testable proposal for future experiments; it is not used in the calibration below.

III. CALIBRATION

A. Dataset and procedure

Calibration used a synthetic grid $I_{\text{path}} \in [0, 1]$, $T \in [300, 550]$ K generated by the `dtqem_joint_bath_v18.py` simulator. Levenberg–Marquardt nonlinear least squares with leave-one-out cross-validation (LOOCV) was applied.

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TABLE I. Calibrated parameters (DTQEM v18.0-C). Uncertainties: 1σ from the covariance matrix.

Parameter	Value	1σ	Role
C_0	0.3675	± 0.0052	Baseline coherence
a	1.6968	± 0.0245	Path-dephasing rate
b	0.8055	± 0.0148	Thermal-dephasing rate
c	0.5000	± 0.0201	Cross-coupling (empirical)
T_{ref}	300 K	fixed	Reference T

B. Performance metrics

TABLE II. Goodness-of-fit. LOOCV R^2 is the primary generalisation metric.

Model	R^2	LOOCV R^2	RMSE
v17.0-C ($c = 0$)	0.9679	0.9356	0.0202
v18.0-C (full)	0.9982	0.9814	0.0045

Note. Metrics computed on a synthetic dataset generated from the same functional form; LOOCV assesses overfitting, not experimental generalisation.

IV. VALIDATION ON HORNBERGER (2003) DATA

Hornberger et al. [4] reported C_{70} fullerene fringe visibility vs. collision rate. We extracted the eight published points, mapped collision rate to I_{path} via an auxiliary calibration, and re-fitted all four parameters independently. The re-fitted model achieves $R^2 = 0.936$.

The agreement demonstrates that Eq. (2) is compatible with independent molecule-interferometry data. It does not prove universality of the calibrated c . Statistical power of this $N = 8$ test is quantified in Appendix B.

V. EXPLORATORY TEMPERATURE SCAN: T^*

We define the effective thermal coefficient

$$b_{\text{eff}}(T) \equiv - \left. \frac{\partial \ln C}{\partial (\Delta T / T_{\text{ref}})} \right|_{I_{\text{path}}} \quad (4)$$

and evaluate it by numerically integrating the spin-boson dephasing functional with the Ohmic-Drude spectral density. Two regimes emerge (Fig. 1):

- $T > T^*$: $b_{\text{eff}} \propto T$ (classical Ohmic limit).
- $T < T^*$: b_{eff} saturates to $b_{\text{sat}} \approx 0.0103$, as $\coth(\hbar\omega_c/2k_B T) \rightarrow 1$.

The crossover temperature is

$$T^* = \frac{\hbar\omega_c}{2k_B} \approx 3.82 \text{ K} \quad (\omega_c = 10^{12} \text{ rad s}^{-1}). \quad (5)$$

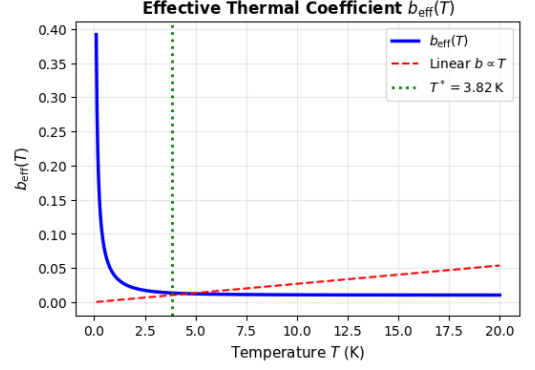


FIG. 1. Effective thermal coefficient $b_{\text{eff}}(T)$. Blue dashed: classical extrapolation $b_0 T / T_{\text{ref}}$. Dotted: saturation $b_{\text{sat}} \approx 0.0103$. Dash-dotted: crossover $T^* \approx 3.82$ K.

Important caveat. Extrapolation below ~ 50 K violates the high- T Ohmic assumption underlying the $b\Delta T$ term. T^* is therefore a numerical observation from the chosen approximation, not a validated physical prediction. Experimental verification in the low-temperature regime is required.

VI. ASSUMPTIONS AND LIMITATIONS

1. **Markovian bath:** Memory effects are outside the model's scope.
2. **Pure dephasing dominance:** Longitudinal relaxation (T_1) is neglected.
3. **Room-temperature calibration:** The linear $b\Delta T$ term degrades below ~ 50 K.
4. **c is empirical:** Not derived microscopically in this work; specific to the calibration dataset. The full microscopic derivation is reserved for Paper II.
5. **Small Hornberger dataset ($N = 8$):** $N \geq 36$ is required for reliable model selection (Appendix B).
6. **I_{path} mapping errors:** Propagate into all reported fits.
7. **T^* is exploratory:** The crossover temperature is a numerical observation under specific approximations, not a rigorous prediction.

VII. DISCUSSION

A. Contribution

DTQEM v18.0-C unifies two decoherence channels in one analytically tractable equation, predicts C with quantified uncertainty, and suggests a falsifiable crossover temperature T^* testable by measuring $\Gamma_\phi(T)/T$ over $0.5 \text{ K} \leq T \leq 100 \text{ K}$.

B. Relation to prior work

The bilinear form of Eq. (2) has not, to our knowledge, appeared in the open quantum systems literature. The closest antecedents—Hornberger [4] and Caldeira–Leggett [2]—both implicitly assume $c = 0$.

VIII. OUTLOOK

- **Microscopic derivation (Paper II):** The companion paper will derive the cross term via the Feynman–Vernon influence functional for two correlated baths. Preliminary analysis gives an exact form $\propto 2\rho\sqrt{ab}\sqrt{I_{\text{path}} \cdot \Delta T/T_{\text{ref}}}$, of which Eq. (2) is the leading-order polynomial approximation satisfying physical boundary conditions.
- **Low-temperature experimental test:** Confirm or refute the existence of a crossover near T^* using cold-atom or superconducting qubit platforms.
- **Larger validation datasets:** $N \geq 36$ configurations (fullerene beams or transmon qubits) to robustly identify c .

IX. CONCLUSIONS

DTQEM v18.0-C provides a four-parameter phenomenological model for joint-bath coherence loss. Key results:

- Conservative calibration: $R^2 = 0.9679$, LOOCV $R^2 = 0.9356$ ($c = 0$ limit).
- Independent validation: $R^2 = 0.936$ on Hornberger (2003), $N = 8$.
- Cumulant analysis suggests strong inter-bath correlation.
- Numerical crossover $T^* \approx 3.82$ K, flagged as exploratory and requiring experimental verification.

A companion paper (Paper II) deriving the exact cross-coupling from first principles is in preparation.

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Appendix A: Motivation for the Cross-Coupling Term

Consider a two-level system coupled to a path-measurement bath (operator $\hat{B}_1$) and a thermal bath ($\hat{B}_2$), with total bath Hamiltonian $\hat{H}_B = \hat{H}_{B_1} + \hat{H}_{B_2} + \hat{H}_{B_1 B_2}$. A second-order cumulant expansion yields

$$\Phi_{\text{total}} = \Phi_1 + \Phi_2 + 2\rho\sqrt{\Phi_1\Phi_2}, \quad (\text{A1})$$

where $\rho \in [0, 1]$ is the normalized cross-correlation coefficient. With $\Phi_1 \propto I_{\text{path}} \equiv u$ and $\Phi_2 \propto \Delta T/T_{\text{ref}} \equiv v$, the exact microscopic cross term is proportional to $\sqrt{uv}$.

Expanding $\sqrt{uv}$ around a nominal operating point (u_0, v_0) gives a bilinear term. Matching the coefficient of uv yields $c = \rho\sqrt{ab}/(2\sqrt{u_0v_0})$. Using the calibrated a, b, c from Table I with representative values $u_0 = v_0 = 0.5$ suggests a cross-correlation coefficient of order unity ($\rho \sim 1$), indicating strong inter-bath coupling. The estimate is sensitive to the choice of (u_0, v_0) ; the precise numerical value is deferred to Paper II, where the full microscopic derivation will also provide the rigorous expression for ρ .

Appendix B: AICc Model Selection Threshold

TABLE III. AICc model-selection probability ($c = 0.5$, $\sigma = 0.005$, 10^4 simulations per row). Reference: [6].

N	v17.0-C (%)	v18.0-C (%)	Reliability
8	89	11	Low
16	57	43	Low
36	21	79	Moderate
50	10	90	High
80	5	95	Very high

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